Day 11 – From Annuities to Life-Contingent Insurance Pricing

OSALP Phase 3 Student Learning Slides

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Nurturing Future Risk Managers and Actuaries

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Today we learn how actuaries price promises whose value depends on time, interest, and sometimes life status.

How to use this slide deck

- This lesson begins with a familiar question: How much is a future payment worth today?
- We then extend this idea from one payment to many payments, and from certain payments to payments depending on survival or death.
- The main storyline is:

$$\text{present value} \longrightarrow \text{annuities} \longrightarrow \text{life tables} \longrightarrow \text{life-contingent pricing.}$$

- We do not rely on calculus. We use timelines, tables, plain language, and spreadsheet-style row logic.
- You are encouraged to use Excel, Google Sheets, Google Gemini, and optional R demonstrations to check calculations and explain your reasoning.

Technology Check

Recommended tools today: Excel or Google Sheets, Google Gemini, optional R teacher demonstrations, and printed life-table snippets.

Opening Problem: similar cash amounts, different prices

Cognitive Trigger / Opening Problem

A grandmother wants to create a scholarship fund. It will pay a student HK$12,000 at the end of each year for 5 years.
Another family wants a life insurance policy that pays money only if a parent dies within the next 5 years.
A third person is worried about living a very long life and wants retirement income that continues while they are alive.

Questions for today:

1. How much money should be set aside today for the scholarship fund?
2. Why does the answer change if the payments are made at the beginning of each year?
3. Why can two promises with the same dollar amount have different prices when life and death are involved?
4. Why do life insurance and life annuities protect against opposite financial risks?

Why this topic is meaningful

- Many real financial promises are not just about how much money is paid.
- They also depend on when the money is paid and what event triggers the payment.
- Scholarships, loans, pensions, retirement income plans, insurance benefits, and savings products all use the same basic idea.
- A life table helps us turn uncertain survival and death outcomes into a structured table that can be used for pricing.

Key Definition / Result

Core message of Day 11:

Value today depends on timing.

For life-contingent products, value also depends on

whether a person is alive or has died when the payment date arrives.

Lesson roadmap

1. Money has a time-coordinate: present value and future value.
2. From one payment to a stream of payments.
3. Certain annuities: annuity-immediate and annuity-due.
4. Insurance and annuity product language.
5. Life tables, survival probabilities, death probabilities, and life expectancy.
6. Pricing as:
$$\text{cash flow} \times \text{discount factor} \times \text{probability of payment.}$$
7. Worked examples: temporary life annuity, guaranteed annuity, deferred guaranteed annuity, term insurance, premiums, endowment insurance, and combined policies.
8. Spreadsheet, Google Gemini, and optional R workflows.

Section A. Money has a time-coordinate

- A dollar is not just a dollar. It also has a date attached to it.
- Money available now can be spent, saved, invested, or used to reduce risk.
- Because of this, HK$100 today is usually worth more than HK$100 one year later.
- Actuarial science begins by placing every cash flow on a timeline.

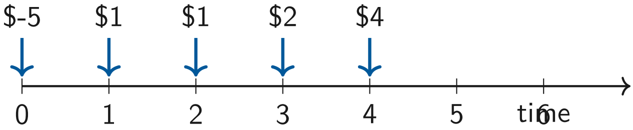
Key Definition / Result

Value measured *today* is called present value (PV).
Value measured at a future date is called future value (FV).

Cash flow: the unit of financial storytelling

- A cash flow is an amount of money received or paid at a particular time point.
- We may use a positive sign for money received and a negative sign for money paid out.
- In actuarial science, the key question is not only "how much money?"
- We also ask:

How much? When? Under what condition?



Key Definition / Result

The cash-flow diagram is the visual bridge from ordinary finance to insurance pricing.

Discounting as a time elevator

- Imagine money travelling in a lift between floors of time.
- Going up from today to the future usually multiplies by a growth factor.
- Coming down from the future back to today usually multiplies by a discount factor.
- The interest rate tells us how strongly the lift changes the value.

Key Definition / Result

If the annual effective interest rate is i, then the one-year discount factor is

$$v = \frac{1}{1+i}.$$

Simple interest versus compound interest

Simple interest

Interest is calculated only on the original principal:

$$a(t) = 1 + it.$$

Compound interest

Interest is reinvested and earns further interest:

$$a(t) = (1 + i)^t.$$

Key Definition / Result

Unless stated otherwise, we use annual compound interest and discount by

$$v^t = (1 + i)^{-t}.$$

The basic symbols of interest theory

- i: annual interest rate.
- v: one-year discount factor.
- n: number of years or periods.
- C_t: cash flow paid at time t.
- $a(t)$: accumulation function from time 0 to time t.

Key Definition / Result

Two basic formulas:

$$FV = PV(1+i)^n, \qquad PV = FVv^n = \frac{FV}{(1+i)^n}.$$

No calculus is needed. We only need repeated multiplication or a spreadsheet.

Worked Example 1: one future payment

Worked Example

A prize of HK$10,000 will be paid in 3 years. The annual effective interest rate is 4%. Find the present value.

1. One-year discount factor:
$$v = \frac{1}{1.04}.$$
2. Three-year discount factor:
$$v^3 = \frac{1}{1.04^3}.$$
3. Present value:
$$PV = 10,000v^3.$$
4. Numerically,
$$PV = \frac{10,000}{1.04^3} \approx \mathbf{8,889.96}.$$

Interpretation: HK$8,889.96 today grows to HK$10,000 in 3 years at 4%.

Technology Check: one future payment

Technology Check

Spreadsheet formula:

```
=10000/(1+0.04)^3
```

Google Gemini prompt:

```
Write an Excel formula for the present value of 10000 received
in 3 years at 4% annual interest. Explain each symbol in plain
language. No calculus.
```

Optional R line:

```r
10000 / (1 + 0.04)^3
```

From one payment to many payments

- Real financial products usually pay more than once.
- A pension, a rent contract, a savings plan, or an insurance income stream involves a sequence of cash flows.
- The present value is found by discounting each payment separately and then adding.

Key Definition / Result

For payments $C_1, C_2, \ldots, C_n$ at times $1, 2, \ldots, n$,

$$PV = \sum_{t=1}^{n} C_t v^t.$$

The sigma sign means add all the rows. It is a compact way of writing a table.

A PV table is often clearer than a long formula

Time t	Cash flow C_t	Discount v^t	Contribution $C_t v^t$	Running total
1	payment at year 1	v	row 1 PV	total so far
2	payment at year 2	v^2	row 2 PV	total so far
3	payment at year 3	v^3	row 3 PV	total so far

- A table makes the time index visible.
- It also matches the way we build models in Excel or Google Sheets.
- This row-by-row thinking will become extremely useful when payments depend on survival or death.

Worked Example 2: a small payment stream

Worked Example

A club promises HK$5,000 at the end of each year for 4 years. The annual interest rate is 5%.

1. Payment dates are $1, 2, 3, 4$.
2. Discount each payment:
$$5000v, \quad 5000v^2, \quad 5000v^3, \quad 5000v^4.$$
3. Add the values:
$$PV = 5000(v + v^2 + v^3 + v^4).$$
4. Numerical value:
$$PV \approx \mathbf{17{,}729.76}.$$

Interpretation: this is the single amount today that is financially equivalent to the four future end-of-year payments.

Exercise: valuing a mixed cash-flow pattern

Investigation / Activity

Consider the cash-flow pattern

$$C_0 = 2, \qquad C_1 = 1, \qquad C_2 = 0, \qquad C_3 = 2, \qquad C_4 = 4,$$

with interest rate 2% per year.

1. What is v?
2. What are $a(t)$ for $t = 0, 1, 2, 3, 4$ under compound interest?
3. What is the present value of the cash flows?
4. What is the value of the same cash flows at time 3?

Key Definition / Result

This exercise trains us to move values between different time points, not just use one standard present value formula.

Section B. What is an annuity?

- An annuity is a sequence of payments made at regular time intervals.
- The payments may be yearly, monthly, or on another fixed schedule.
- We first study level yearly payments.
- Later, we study life-contingent annuities, where payments continue only if a person is alive, except when there is a guarantee.

Key Definition / Result

Examples include tuition support, pension income, rent, loan repayments, retirement income plans, and insurance income streams.

Annuity-immediate versus annuity-due

Annuity-immediate

Payments occur at the end of each period:

$$1, 2, 3, \ldots, n.$$

Annuity-due

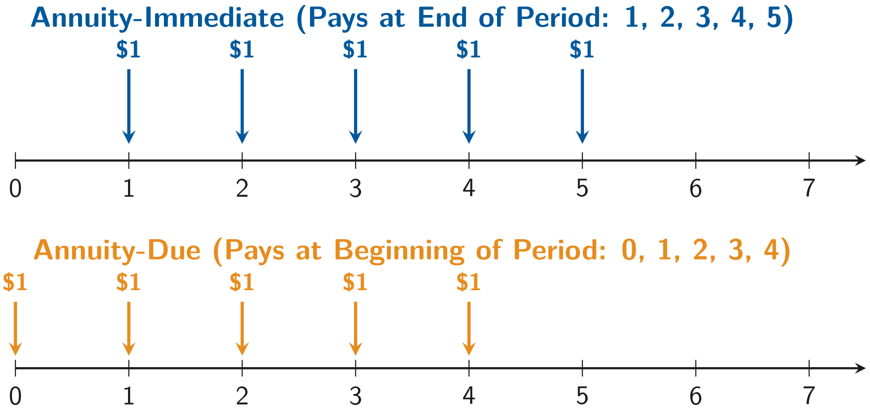
Payments occur at the beginning of each period:

$$0, 1, 2, \ldots, n-1.$$

Key Definition / Result

Same payment amount, same interest rate, and same number of payments can still give different present values because the timing is different.

Timeline picture: immediate versus due



- The annuity-due arrows stand one step earlier.
- Earlier payments are discounted for fewer years. Therefore, the annuity-due is usually worth more than the corresponding annuity-immediate.

Key Definition / Result

If each payment is 1 unit, then for an annuity-immediate,

$$a_{\overline{n}|} = v + v^2 + \cdots + v^n.$$

For an annuity-due,

$$\ddot{a}_{\overline{n}|} = 1 + v + v^2 + \cdots + v^{n-1}.$$

- The dots over a remind us that the payments are made at the beginning of periods.
- If each payment is P, multiply the annuity value by P.
- If you are unsure, return to the timeline and table.

Worked Example

Find the present value of a 5-year annuity-immediate paying HK$8,000 each year at 6%.

1. Payment dates are $1, 2, 3, 4, 5$.
2. Discount each payment:
$$8000v, \quad 8000v^2, \quad \ldots, \quad 8000v^5.$$
3. Add all five discounted values.
4. Numerical value:
$$PV \approx \mathbf{33{,}699.23}.$$

Interpretation: this is the value today of five equal end-of-year payments.

Worked Example 4: the same cash flow as an annuity-due

Worked Example

Now move the same HK$8,000 payment stream to the beginning of each year for 5 years.

1. Payment dates are $0, 1, 2, 3, 4$.
2. Discount factors are now
$$1, \quad v, \quad v^2, \quad v^3, \quad v^4.$$
3. Add the five contributions.
4. Numerical value:
$$PV \approx \mathbf{35{,}721.18}.$$

Comparison: the annuity-due is worth more because each payment arrives earlier.

A shortcut relationship

Key Definition / Result

Because the annuity-due shifts each payment one period earlier,

$$\ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|}.$$

- This formula is a shortcut, not the starting point.
- The timeline explains why it is true.
- If you forget the shortcut, use the row-by-row PV table.

Common mistake: mixing up the payment time

Common Mistake / Pitfall

Students often use the correct cash amount but the wrong set of time points.

Typical mistakes:

- Treating an annuity-due as if the first payment is at time 1.
- Using n payments but accidentally discounting them over $n+1$ dates.
- Writing a spreadsheet where the first discount factor should be 1, but entering v instead.

Section C. Why annuities become actuarial

- In ordinary annuities, the payment schedule is usually certain once the contract is created.
- In life insurance and life annuities, payments depend on life status: alive or dead at a particular time.
- Actuaries therefore combine financial mathematics with survival and death probabilities.

Key Definition / Result

Life insurance protects against having a short life.
Life annuities protect against having a long life.
These are opposite financial risks.

Life insurance versus life annuity

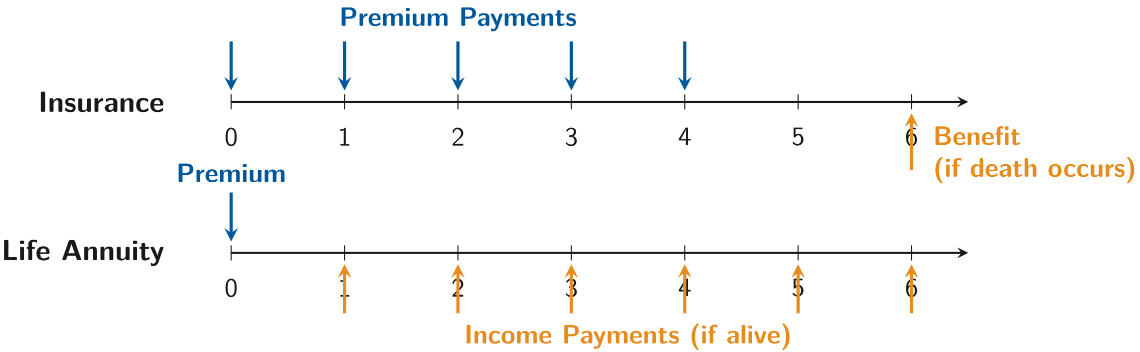
Life insurance

- The policyholder pays premiums to the insurer.
- A death benefit is paid at or after death, depending on the contract.
- Main protection: the family may suffer financially if the insured person dies too early.

Life annuity

- A person pays a premium or accumulated savings to buy future income.
- Payments are received while alive, sometimes with guarantees.
- Main protection: the retiree may live longer than expected and outlive savings.

Timeline interpretation: insurance and annuity



Key Definition / Result

A timeline is safer than a memorised formula because it tells us which event triggers each payment.

Versions of life insurance

- Whole life insurance (終身壽險): a death benefit is paid when death occurs, subject to the exact contract details.
- Term life insurance (定期壽險): a death benefit is provided only for a fixed number of years.
- Endowment insurance (儲蓄保險): a death benefit is provided during a fixed term, and a maturity payment is made if the policyholder survives to the end.
- Pure endowment: a benefit is paid only if the policyholder is alive at a specified future time.

Key Definition / Result

For students, the product name is less important than the payment condition:

death-triggered, survival-triggered, or both.

Versions of life annuity

- Whole life annuity (終身年金): payments continue until death.
- Temporary life annuity (限期年金): payments are made only for a fixed number of years, while the required survival condition is satisfied.
- Deferred annuity (遞延年金): payments begin only after a waiting period.
- Guaranteed payment (付款保證): payments during the guaranteed period are made whether or not the annuitant is alive, subject to the contract wording.

Key Definition / Result

Guaranteed payments reduce the fear of paying a premium and dying very soon, but they also change the price.

Insurance conventions used in this lesson

Common Mistake / Pitfall

Unless a question says otherwise, we use this convention:

- Premiums are paid at the beginning of the year.
- Benefits are paid at the end of the year.

- This is why premium terms and benefit terms often use different time exponents.
- It is one of the most common sources of mistakes in life-contingent pricing.
- Always mark premium dates and benefit dates before writing formulas.

Section D. Mortality tables are structured survival data

- A mortality table records how a large group of lives changes with age.
- Instead of treating life and death uncertainty as completely mysterious, we organise it into a table.
- The table helps us estimate survival and death probabilities for life-contingent payments.

Key Definition / Result

A life table turns uncertainty into rows that we can read, calculate, and discuss.

A life table as a stadium of seats

- Imagine a stadium filled with people all aged x.
- The number still seated one year later is like l_{x+1}.
- The number who left during that year is like d_x.
- The fractions of seats remaining or emptied become survival and death probabilities.

Key Definition / Result

This metaphor helps us read the symbols without needing advanced probability theory.

Core mortality symbols

Key Definition / Result

For age x:

$$\begin{aligned} l_x &:= \text{number alive at exact age } x, \\ d_x &:= \text{number dying between ages } x \text{ and } x+1, \\ q_x &:= \text{probability of death within one year after age } x, \\ p_x &:= \text{probability of surviving one year after age } x. \end{aligned}$$

- The letter l reminds us of lives.
- The letter d reminds us of deaths.
- The probabilities p_x and q_x are calculated from the table.

Life table example

Age x	l_x	d_x
0	100,000	2,000
1	98,000	1,500
2	96,500	1,000
3	95,500	900
$\vdots$	$\vdots$	$\vdots$

Key Definition / Result

Basic relationship:

$$l_{x+1} = l_x - d_x.$$

The table is just counting, but those counts become probabilities after we divide by the correct starting number.

Basic life-table relationships

Key Definition / Result

The table relations are

$$d_x = l_x - l_{x+1}, \qquad q_x = \frac{d_x}{l_x}, \qquad p_x = \frac{l_{x+1}}{l_x} = 1 - q_x.$$

- These formulas only use differences and ratios.
- They are ideal for spreadsheet formulas.
- The most important habit is to use the correct denominator.

Probabilities from the life table example

Worked Example

Using $l_0 = 100,000$, $l_1 = 98,000$, $l_2 = 96,500$, and $l_3 = 95,500$:

1. $$ {}_2p_0 = \frac{l_2}{l_0} = \frac{96,500}{100,000} = 0.965. $$
2. $$ {}_2q_1 = 1 - \frac{l_3}{l_1} = 1 - \frac{95,500}{98,000}. $$
3. $$ p_1 = \frac{l_2}{l_1} = \frac{96,500}{98,000}. $$
4. $$ q_0 = \frac{d_0}{l_0} = \frac{2,000}{100,000} = 0.02. $$

Key Definition / Result

Multi-year survival notation

Key Definition / Result

The symbol ${}_np_x$ means:

probability that a life aged x survives for n more years.

The symbol ${}_nq_x$ means:

probability that a life aged x dies within the next n years.

- The small n acts like a travel distance marker.
- For example, ${}_2p_0$ means "from age 0, survive two years".

How to compute ${}_np_x$ from a table

Key Definition / Result

Two equivalent ways:

$$ {}_np_x = p_x p_{x+1} \cdots p_{x+n-1} $$

and, when the l_x values are available,

$$ {}_np_x = \frac{l_{x+n}}{l_x}. $$

- The product formula shows the year-by-year survival logic.
- The l_x formula is often faster and clearer in a spreadsheet.

Life expectancy: plain-language meaning

- Life expectancy asks:

  How many additional complete years can a person aged x expect to live on average?

- Let Y be the number of additional complete years lived by a person already aged x.
- Then $e_x = E(Y)$ is the expected value of that future lifetime count.

Key Definition / Result

In this introductory lesson, life expectancy is mainly a conceptual extension of life-table probabilities.

Life expectancy formula from survivor counts

Key Definition / Result

A useful discrete formula is

$$e_x = \frac{1}{l_x} \sum_{k=1}^{\infty} l_{x+k}.$$

- The formula says: add future survivor counts, then divide by the number alive now.
- For secondary students, the key interpretation is more important than the notation.
- Higher future survivor counts lead to a higher expected number of additional complete years.

Challenge: life table and expectancy

Investigation / Activity

Let

$$q_{60} = 0.20, \quad q_{61} = 0.25, \quad q_{62} = 0.25, \quad q_{63} = 0.30, \quad q_{64} = 0.40.$$

1. Find l_x for ages 60 to 65, beginning with $l_{60} = 1000$.
2. Find the probability that a 61-year-old will die between the ages 62 and 64.
3. Find the probability that a 62-year-old will live to age 65.

Solution structure for the challenge I

Worked Example

Start from $l_{60} = 1000$ and use

$$l_{x+1} = l_x(1 - q_x).$$

Age	q_x	l_x
60	0.20	1000
61	0.25	800
62	0.25	600
63	0.30	450
64	0.40	315
65	–	189

Solution structure for the challenge II

- A 61-year-old dies between ages 62 and 64 if they survive from 61 to 62 and then die before 64:

$$ {}_1p_{61\,2}q_{62} = 0.75\left(1 - \frac{l_{64}}{l_{62}}\right) = 0.75(1 - 315/600) = 0.35625. $$

- A 62-year-old lives to age 65:

$$ {}_3p_{62} = \frac{l_{65}}{l_{62}} = \frac{189}{600} = 0.315. $$

Backward recursion idea for life expectancy

Key Definition / Result

A student-friendly recursion is

$$e_x = p_x(1 + e_{x+1}).$$

It says: to enjoy one more complete year and the future expectation after that, the life must first survive the next year.

Worked Example

Given $e_{64} = 0.8$:

$$\begin{aligned}
e_{63} &= 0.70(1.8) = 1.26, \\
e_{62} &= 0.75(2.26) = 1.695, \\
e_{61} &= 0.75(2.695) = 2.02125, \\
e_{60} &= 0.80(3.02125) = 2.417.
\end{aligned}$$

Section E. Pricing life-contingent products

Key Definition / Result

Every pricing example today follows one row logic:

$$\text{row value} = \text{cash flow} \times \text{discount factor} \times \text{probability that the row is paid.}$$

- For annuity payments, the probability is usually a survival probability.
- For death benefits, the probability is usually:

$$\text{survive to the beginning of the year} \times \text{die during the year.}$$

- Premiums are also cash flows, so we can set the present value of premiums equal to the present value of benefits.

The one key idea for pricing

Life annuity price

$$\text{Annuity price} = \sum_{t \geq 0} a^{-1}(t) c(t) \, {}_t p_x.$$

Payment at time t is weighted by the probability of being alive to receive it.

Life insurance price

$$\text{Insurance price} = \sum_{t \geq 0} a^{-1}(t+1) c(t+1) \, {}_t p_x q_{x+t}.$$

Benefit at the end of year $t+1$ is weighted by surviving to the start of that year and dying during it.

Key Definition / Result

You do not need to memorise these formulas. Read each row as:

$$\text{discount factor} \times \text{cash flow} \times \text{probability of payment}.$$

Worked Example 5: a simple life-contingent value

Worked Example

A benefit of HK\$20,000 is paid at the end of year 3 if a life aged 60 survives 3 years. Assume $i = 5\%$ and ${}_3p_{60} = 0.9667$.

1. Discount the amount to today:
$$20,000v^3.$$
2. Multiply by the survival probability.
3. Thus,
$$\text{value} = 20,000 \times \frac{1}{1.05^3} \times 0.9667.$$
4. Numerical value:
$$\text{value} \approx \mathbf{16{,}704}.$$

Interpretation: the payment is valuable, but less than a certain HK\$20,000 in year 3 because survival is not guaranteed.

Worked Example 6: a level life annuity in table language

Worked Example

Suppose HK$5,000 is paid at the end of each of the next 3 years only while the person is alive.

Year	Payment if alive	Survival probability	Discount factor	Weighted PV row
1	5000	${}_1p_x$	v	$5000({}_1p_x)v$
2	5000	${}_2p_x$	v^2	$5000({}_2p_x)v^2$
3	5000	${}_3p_x$	v^3	$5000({}_3p_x)v^3$

Add the rows to get the actual present value.

Worked Example 7: temporary life annuity with variable payments

Worked Example

An aged 50 person is provided with payments of \$1 at age 50, \$2 at age 51, \$3 at age 52, and \$4 at age 53.
Suppose

$$q_{50} = 0.1, \qquad q_{51} = 0.2, \qquad q_{52} = 0.25.$$

The interest rate is 10% for the first year and 20% thereafter. Find the price.

Key Definition / Result

This example is useful because the payments are not level and the interest rate changes after the first year.

Worked Example

The survival probabilities are

$$ {}_0p_{50} = 1, \qquad {}_1p_{50} = 1 - q_{50} = 0.9, $$
$$ {}_2p_{50} = (1 - q_{50})(1 - q_{51}) = 0.9(0.8), $$
$$ {}_3p_{50} = (1 - q_{50})(1 - q_{51})(1 - q_{52}) = 0.9(0.8)(0.75). $$

Therefore,

$$ \text{Price} = 1 + \frac{2(0.9)}{1.1} + \frac{3(0.9)(0.8)}{1.1(1.2)} + \frac{4(0.9)(0.8)(0.75)}{1.1(1.2)^2}. $$

$$ \text{Price} \approx \mathbf{5.909}. $$

Worked Example 8: guaranteed annuity

Worked Example

A person aged 40 purchases a life annuity that provides \$10,000 each year for life, with the first payment starting at age 41. The first 10 payments will be paid regardless of whether the annuitant is alive or not. Find a formula for the premium.

Key Definition / Result

The first 10 payments are certain. Payments after the guaranteed period are life-contingent.

$$\text{Premium} = 10,000 \left(\sum_{t=1}^{10} v^t + \sum_{t=11}^{\infty} v^t \, {}_t p_{40} \right).$$

Worked Example 9: deferred plus guaranteed annuity

Worked Example

A person aged 40 purchases a life annuity that provides \$10,000 each year for life, with the first payment starting at age 65. If he reaches age 65, he will receive at least 10 payments regardless of whether he is alive after age 65. Find a formula for this premium.

Key Definition / Result

This combines three ideas:

deferment to age 65, survival to the start of payment, a 10-payment guarantee.

$$\text{Premium} = 10,000\,{}_{25}p_{40}\left(\sum_{j=25}^{34} v^j + \sum_{j=35}^{\infty} v^j\,{}_{j-25}p_{65}\right).$$

Worked Example 10: term life insurance

Worked Example

An aged 60 person is provided \$800 payable at the end of the year of death in the first year, \$750 in the second year, and \$1000 in the third year. Nothing is paid if the person survives after age 63.

Suppose

$$q_{60} = 0.2, \qquad q_{61} = 0.4, \qquad q_{62} = 0.5, \qquad i = 10\%.$$

What is the price?

Key Definition / Result

This is a death-benefit problem, so each row uses

survive to the start of the policy year $\times$ die during that policy year.

Solution: term life insurance

Worked Example

The three payment probabilities are

$$q_{60}, \qquad {}_1p_{60}\,q_{61}, \qquad {}_2p_{60}\,q_{62}.$$

Thus,

$$\text{Price} = \frac{800(0.2)}{1.1} + \frac{750(0.8)(0.4)}{1.1^2} + \frac{1000(0.8)(0.6)(0.5)}{1.1^3}.$$

$$\text{Price} \approx \mathbf{524.1}.$$

- The first term pays if death occurs in the first year.
- The second term pays if the person survives one year and dies in the second year.
- The third term pays if the person survives two years and dies in the third year.

Worked Example 11: solving for premiums

Worked Example

Use the same term insurance as before, with benefit present value \$524.1. The insurance is purchased by paying

$$P \text{ at age } 60, \qquad 2P \text{ at age } 61, \qquad 0.75P \text{ at age } 62,$$

at the beginning of each year, provided the policyholder is alive at those payment times. Find P.

Key Definition / Result

Premiums are also life-contingent cash flows.
The equivalence principle says:

$$\text{PV of premiums} = \text{PV of benefits}.$$

Solution: solving for premiums

Worked Example

The premium present value is

$$P + \frac{2P(0.8)}{1.1} + \frac{0.75P(0.8)(0.6)}{1.1^2}.$$

Set this equal to the benefit value:

$$P + \frac{2P(0.8)}{1.1} + \frac{0.75P(0.8)(0.6)}{1.1^2} = 524.1.$$

Solving gives

$$P \approx \mathbf{190.4}.$$

Common Mistake / Pitfall

Do not discount the first premium by v. It is paid at the beginning, at time 0.

Worked Example 12: endowment insurance I

Worked Example

Consider a 20-year endowment insurance with death benefit equal to 1 unit for the first 10 years, and 2 units for the second 10 years. The pure endowment amount is 3 units. Level premiums of P are payable for 15 years. Find P.

Key Definition / Result

An endowment combines death benefits before maturity and a survival benefit at maturity.

$$\text{PV benefits} = \sum_{t=0}^{9} v^{t+1}\,{}_tp_x q_{x+t} + 2\sum_{t=10}^{19} v^{t+1}\,{}_tp_x q_{x+t} + 3v^{20}\,{}_{20}p_x.$$

$$\text{PV premiums} = P\sum_{t=0}^{14} v^t\,{}_tp_x. \quad P = \frac{\text{PV benefits}}{\sum_{t=0}^{14} v^t\,{}_tp_x}.$$

Worked Example

A contract on an aged 40 person provides annuity benefits of 1 per year for life, beginning at age 65. If the person dies before age 65, a death benefit of 10 is paid. Level annual premiums of P are payable for 25 years. Find P.

Key Definition / Result

This question combines:

a deferred life annuity, a temporary death benefit, a premium annuity.

$$\text{PV deferred annuity} = \sum_{t=25}^{\infty} v^t {}_tp_{40}. \quad \text{PV death benefit} = 10\sum_{t=0}^{24} v^{t+1} {}_tp_{40} q_{40+t}.$$

$$\text{PV premiums} = P\sum_{t=0}^{24} v^t {}_tp_{40}. \quad P = \frac{\text{PV deferred annuity} + \text{PV death benefit}}{\sum_{t=0}^{24} v^t {}_tp_{40}}.$$

Technology-first laboratory mindset

- Because students have no calculus background, we choose timelines, tables, spreadsheet rows, and interpretable symbols.
- Technology can handle arithmetic.
- Students should focus on what each row means.
- The goal is not to make the spreadsheet look fancy. The goal is to make each row mathematically meaningful.

Key Definition / Result

If a spreadsheet number looks strange, check the timeline before changing formulas randomly.

Interpretation checklist after every answer

Key Definition / Result

After computing any result, ask five questions:

1. Is this a value today or a value in the future?
2. Is the first payment at the beginning or end of the period?
3. Does survival matter for this payment?
4. Does death during a particular year matter for this benefit?
5. Is the magnitude reasonable compared with the original cash amounts?

Review Set 11A – foundations

1. Define present value in one sentence.
2. Write the one-year discount factor in terms of i.
3. Explain in words why earlier payments are worth more.
4. Distinguish annuity-immediate and annuity-due.
5. What does l_x mean?
6. What does q_x mean?
7. What is the difference between life insurance and life annuity?
8. Why should we still interpret the output rather than trust software blindly?

Review Set 11B – challenge thinking

(1) A student says two 5-payment annuities must have equal PV because the cash totals are equal. Explain why this is false.

(2) Why is a timeline often safer than memorising formulas?

(3) Explain in words how life contingencies extend ordinary present value calculations.

(4) Compare a certain payment in year 4 with a year-4 payment that happens only if someone is alive. Which is worth more, and why?

(5) Compare a death benefit and a survival benefit. What event triggers each payment?

(6) Create a one-row actuarial present value expression for a benefit paid at time 2 only if alive at time 2.

(7) Create a one-row actuarial present value expression for a benefit paid at the end of year 2 only if death occurs during year 2.

Common pitfalls to keep visible

Common Mistake / Pitfall

- Forgetting whether the first payment is at time 0 or time 1.
- Treating an annuity as a single lump sum.
- Mixing up l_x and d_x.
- Reading ${}_np_x$ as a payment amount rather than a survival probability.
- Using ${}_tp_x$ for a death benefit row where the correct probability is ${}_tp_xq_{x+t}$.
- Letting software produce a number without checking whether the timeline makes sense.

Closing learning message

Key Definition / Result

The main modelling habit is simple but powerful:

1. Write the timeline first.
2. Identify the payment amount at each time point.
3. Decide whether each payment is certain, survival-contingent, death-contingent, deferred, guaranteed, or combined.
4. Assign the correct probability of payment.
5. Discount each row to today and add the rows.

If the timeline is clear, the formula becomes much easier to understand.